Types of Graphs

Exam code: 0607

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Written by: Naomi C
Reviewed by: Dan Finlay
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Types of Graphs

What graphs do I need to know?

- You need to be able to **recognise** the following **lines**:
  - **Straight lines**
    - $y = mx + c$
    - Such as $y = 3x + 2$, $y = 5x - 1$, ...
    - Two **important** ones are $y = x$ and $y = -x$
  - **Horizontal lines**
    - $y = c$
    - Such as $y = 4$, $y = -10$, ...
  - **Vertical lines**
    - $x = k$
    - Such as $x = 2$, $x = -1$, ...
- You need to be able to recognise **quadratic** graphs
  - $y = x^2$
  - $y = -x^2$
  - $y = ax^2 + bx + c$

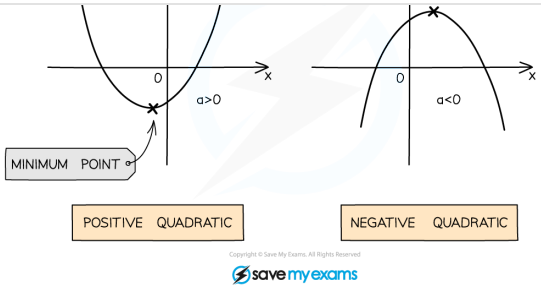
What does a quadratic graph look like?

- The **equation** of a **quadratic graph** is $y = ax^2 + bx + c$
- A quadratic graph has either a **u-shape** or an **n-shape**
  - This type of shape is called a **parabola**
- **u-shapes** are called **positive** quadratics
  - because the number in front of x^2 is positive
    - For example, $y = 2x^2 + 3x + 4$
- **n-shapes** are called **negative** quadratics
  - because the number in front of x^2 is negative
    - For example, $y = -3x^2 + 2x + 4$
- You can plot quadratic graphs using a **table of values**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

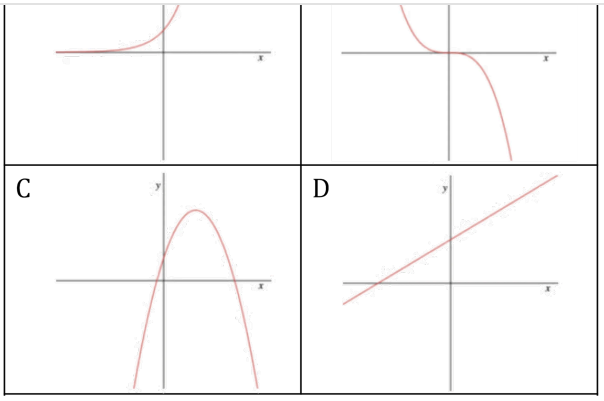


What are the key features of a quadratic graph?

- The point where the graph turns is called the **vertex**
  - **Positive** quadratics have a **minimum** point
    - The bottom of the u-shape
  - **Negative** quadratics have a **maximum point**
    - The top of the n-shape
- Quadratic graphs always have a **vertical line of symmetry** down the **middle**
  - The **equation** of the vertical line of symmetry is $x = k$
    - k is the x-coordinate of the minimum or maximum point
- Quadratic graphs do **not** have to cross the x-axis
  - If they do, **two x-intercepts** are created, called **roots**
    - If the curve just **touches** the x-axis, only **1 root** is created
  - **Roots** are **symmetric** about the **vertical line of symmetry**
- Quadratic graphs **always** have one **y-intercept**

Worked Example

In each of the cases below, state the letter of the graph that corresponds to the equation given.



(a) $y = x + 5$

This is a straight-line graph, $y = mx + c$

Graph D

(b) $y = -x^2 + 3x + 2$

This is a quadratic graph, $y = ax^2 + bx + c$ ($a = -1, b = 3, c = 2$)

The number in front of x^2 is negative so it has an n-shape

Graph C

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